

ASSIGNMENT NO. 8

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ROLL: B-53

Lab Handout 8: Page Replacement Strategies (FIFO, LRU, Optimal)

CODE ->

```
#include <stdio.h> #define MAX

30 int search(int key,int frame[],int f){
for(int i=0;i<f;i++) if(frame[i]==key) return i;
return
-1;
}

int predict(int pages[],int frame[],int n,int index,int f){ int res=-1, farthest=index;
for(int i=0;i<f;i++){ int j; for(j=index;j<n;j++) if(frame[i]==pages[j]){
if(j>farthest){farthest=j; res=i;} break; } if(j==n) return i;
} return (res==-1)?0:res;
}

void simulate(int pages[],int n,int f,char type){ int
frame[MAX], used[MAX], count=0, idx=0; for(int
i=0;i<f;i++) frame[i]=-1; for(int i=0;i<n;i++){ int pos=search(pages[i],frame,f);
if(pos==-1){ if(idx<f){ frame[idx]=pages[i]; used[idx]=i; idx++; } else{
if(type=='F') pos=0; else if(type=='L'){ int min=999; for(int j=0;j<f;j++)
if(used[j]<min){min=used[j];pos=j;} } else pos=predict(pages,frame,n,i,f);
frame[pos]=pages[i]; used[pos]=i;
}
count++;
} else used[pos]=i;
}

printf("%s Page Faults =
%d\n",(type=='F')?"FIFO":(type=='L')?"LRU":"Optimal",count); }
```

```

int main(){  int n,f,pages[MAX];
printf("Enter number of pages: ");
scanf("%d",&n);  printf("Enter page
reference string:\n");  for(int i=0;i<n;i++)
scanf("%d",&pages[i]);  printf("Enter number
of frames: ");  scanf("%d",&f);
simulate(pages,n,f,'F');
simulate(pages,n,f,'L');
simulate(pages,n,f,'O');  return 0;
}

```

Output:

```

Enter number of pages: 12
Enter page reference string:
1 3 0 3 5 6 3 0 3 1 2 1
Enter number of frames: 3
FIFO Page Faults = 8
LRU Page Faults = 8
Optimal Page Faults = 7

```

```

Enter number of pages: 8
Enter page reference string:
7 0 1 2 0 3 0 4
Enter number of frames: 3
FIFO Page Faults = 6
LRU Page Faults = 6
Optimal Page Faults = 6

```

```

Enter number of pages: 20
Enter page reference string:
1 2 3 4 1 2 5 1 2 3 4 5 3 2 4 1 5 3 2 4
Enter number of frames: 4
FIFO Page Faults = 9
LRU Page Faults = 13
Optimal Page Faults = 8

```